

Instrumental Variables & 2SLS

$$\diamond y_1 = b_0 + b_1 y_2 + b_2 z_1 + \dots b_k z_k + u$$

$$\diamond y_2 = p_0 + p_1 z_{k+1} + p_2 z_1 + \dots p_k z_k + v$$

Why Use Instrumental Variables?

- ◆ Instrumental Variables (IV) estimation is used when your model has endogenous x 's
- ◆ i.e. whenever $\text{Cov}(x, u) \neq 0$
- ◆ Thus, IV can be used to address the problem of omitted variable bias
- ◆ Also, IV can be used to solve the classic errors-in-variables problem

What Is an Instrumental Variable?

- ◆ In order for a variable, z , to serve as a valid instrument for x , the following must be true
 1. The instrument must be exogenous
 - i.e. $\text{Cov}(z, u) = 0$
 - more specifically z should have no “partial” effect on y and should be uncorrelated with u
 2. The instrument must be correlated with the endogenous variable x
 - i.e. $\text{Cov}(z, x) \neq 0$

Difference between IV and Proxy?

- ◆ With IV we will leave the unobserved variable in the error term but use an estimation method that recognizes the presence of the omitted variable
- ◆ With a proxy we were trying to remove the unobserved variable from the error term e.g. IQ
- ◆ IQ would make a poor instrument as it would be correlated with the error in our model (ability in u)
- ◆ Need something correlated with education but uncorrelated with ability (parents education?)

More on Valid Instruments

- ◆ We can't test if $\text{Cov}(z,u) = 0$ as this is a population assumption
- ◆ Instead, we have to rely on common sense and economic theory to decide if it makes sense
- ◆ However, we can test if $\text{Cov}(z,x) \neq 0$ using a random sample
- ◆ Simply estimate $x = p_0 + p_1 z + v$, and test $H_0: p_1 = 0$
- ◆ Refer to this regression as the “first-stage regression”

IV Estimation in the Simple Regression Case

- ◆ We can show that if we have a valid instrument we can get consistent estimates of the parameters
- ◆ For $y = b_0 + b_1x + u$
- ◆ $\text{Cov}(z,y) = b_1\text{Cov}(z,x) + \text{Cov}(z,u)$, so
- ◆ Thus, $b_1 = \text{Cov}(z,y) / \text{Cov}(z,x)$, since $\text{Cov}(z,u)=0$ and $\text{Cov}(z,x) \neq 0$
- ◆ Then the IV estimator for b_1 is

$$\hat{b}_1 = \frac{\sum (z_i - \bar{z})(y_i - \bar{y})}{\sum (z_i - \bar{z})(x_i - \bar{x})}$$

◆ Same as OLS
when $z = x$

Inference with IV Estimation

- ◆ The IV estimator also has an approximate normal distribution in large samples
- ◆ To get estimates of the standard errors we need a slightly different homoskedasticity assumption:
 $E(u^2|z) = s^2 = \text{Var}(u)$ (conditioning on z here)
- ◆ If this is true, we can show that the asymptotic variance of β_1 -hat is:

$$\text{Var}(\hat{b}_1) = \frac{s^2}{n s_x^2 r_{x,z}^2}$$

- ◆ σ_x^2 is the pop variance of x
- ◆ σ^2 is the pop variance of u
- ◆ ρ_{xz}^2 is the square of the pop correlation between x and z

Inference with IV Estimation

◆ Each of the elements in the population variance can be estimated from a random sample

◆ The estimated variance is then:

$$Var(\hat{b}_1) = \frac{\hat{s}^2}{SST_x R_{x,z}^2}$$

◆ Where $\sigma^2 = SSR$ from IV divided by the df, SST_x is the sample variance in x and the R^2 is from the first stage regression (x on z)

◆ The standard error is just the square root of this

IV versus OLS estimation

- ◆ Standard error in IV case differs from OLS only in the R^2 from regressing x on z
- ◆ Since $R^2 < 1$, IV standard errors are larger
- ◆ However, IV is consistent, while OLS is inconsistent, when $\text{Cov}(x, u) \neq 0$
- ◆ Notice that the stronger the correlation between z and x , the smaller the IV standard errors

The Effect of Poor Instruments

- ◆ What if our assumption that $\text{Cov}(z,u) = 0$ is false?
- ◆ The IV estimator will be inconsistent also
- ◆ We can compare the asymptotic bias in OLS to that in IV in this case:

$$\text{IV : } \text{plim} \hat{b}_1 = b_1 + \frac{\text{Corr}(z,u)}{\text{Corr}(z,x)} \bullet \frac{s_u}{s_x}$$

$$\text{OLS : } \text{plim} \tilde{b}_1 = b_1 + \text{Corr}(x,u) \bullet \frac{s_u}{s_x}$$

- ◆ Even if $\text{Corr}(z,u)$ is small the inconsistency can be large if $\text{Corr}(z,x)$ is also very small

Effect of Poor Instruments (cont)

- ◆ So, it is not necessarily better to use IV instead of OLS even if z and u are not “highly” correlated
- ◆ Instead, prefer IV only if $\text{Corr}(z,u)/\text{Corr}(z,x) < \text{Corr}(x,u)$
- ◆ Also notice that the inconsistency gets really large if z and x are only loosely correlated
- ◆ Best to test for correlation in the first stage regression

A Note on R^2 in IV

- ◆ R^2 after IV estimation can be negative
- ◆ Recall that $R^2 = 1 - SSR/SST$ where SSR is the residual sum of IV residuals
- ◆ SSR in this case can be larger than SST making the R^2 negative
- ◆ Thus, R^2 isn't very useful here and can't be used for F-tests
- ◆ Not important as we would prefer consistent estimates of the coefficients

IV Estimation in the Multiple Regression Case

- ◆ IV estimation can be extended to the multiple regression case
- ◆ Estimating: $y_1 = b_0 + b_1 y_2 + b_2 z_1 + u_1$
- ◆ Where y_2 is endogenous and z_1 is exogenous
- ◆ Call this the “structural model”
- ◆ If we estimate the structural model the coefficients will be biased and inconsistent
- ◆ Thus, we need an instrument for y_2
- ◆ Can we use z_1 if it is correlated with y_2 (we know it isn't correlated with u_1)?

Multiple Regression IV (cont)

- ◆ No, because it appears in the structural model
- ◆ Instead, we need an instrument, z_2 , that:
 1. Doesn't belong in the structural model
 2. Is uncorrelated with u_1
 3. Is correlated with y_2 in a particular way
 - Now because of z_1 we need a partial correlation
 - i.e. for the “reduced form equation”
$$y_2 = p_0 + p_1 z_1 + p_2 z_2 + v_2, p_2 \neq 0$$
- ◆ If we have such an instrument and u_1 is uncorrelated with z_1 the model is “identified”

Two Stage Least Squares (2SLS)

- ◆ It is possible to have multiple instruments
- ◆ Consider the structural model, with 1 endogenous, y_2 , and 1 exogenous, z_1 , RHS variable
- ◆ Suppose that we have two valid instruments, z_2 and z_3
- ◆ Since z_1 , z_2 and z_3 are uncorrelated with u_1 , so is any linear combination of these
- ◆ Thus, any linear combination is also a valid instrument

Best Instrument

- ◆ The best instrument is the one that is most highly correlated with y_2
- ◆ This turns out to be a linear combination of the exogenous variables
- ◆ The reduced form equation is:
$$y_2 = p_0 + p_1 z_1 + p_2 z_2 + p_3 z_3 + v_2 \text{ or } y_2 = y_2^* + v_2$$
- ◆ Can think of y_2^* as the part of y_2 that is uncorrelated with u_1 and v_2 as the part that might be correlated with u_1
- ◆ Thus the best IV for y_2 is y_2^*

More on 2SLS

- ◆ We can estimate y_2^* by regressing y_2 on z_1 , z_2 and z_3 – the first stage regression
- ◆ If then substitute $\hat{y}_2$ for y_2 in the structural model, get same coefficient as IV
- ◆ While the coefficients are the same, the standard errors from doing 2SLS by hand are incorrect
- ◆ Also recall that since the R^2 can be negative F-tests will be invalid
- ◆ Stata will calculate the correct standard error and F-tests

More on 2SLS (cont)

- ◆ We can extend this method to include multiple endogenous variables
- ◆ However, we need to be sure that we have at least as many excluded exogenous variables (instruments) as there are endogenous variables
- ◆ If not, the model is not identified

Addressing Errors-in-Variables with IV Estimation

- ◆ Recall the classical errors-in-variables problem where we observe x_1 instead of x_1^*
- ◆ Where $x_1 = x_1^* + e_1$, we showed that when x_1 and e_1 are correlated the OLS estimates are biased
- ◆ We maintain the assumption that u is uncorrelated with x_1^* , x_1 and x_2 and that e_1 is uncorrelated with x_1^* and x_2
- ◆ If we can find an instrument, z , such that $\text{Corr}(z, u) = 0$ and $\text{Corr}(z, x_1) \neq 0$, then we can use IV to remove the attenuation bias

Example of Instrument

◆ Suppose that we have a second measure of $x_1^*(z_1)$

Examples:

◆ both husband and wife report earnings

◆ both employer and employee report earnings

◆ z_1 will also measure x_1^* with error

◆ However, as long as the measurement error in z_1 is uncorrelated with the measurement error in x_1 , z_1 is a valid instrument

Testing for Endogeneity

- ◆ Since OLS is preferred to IV if we do not have an endogeneity problem, then we'd like to be able to test for endogeneity
- ◆ Suppose we have the following structural model:
$$y_1 = b_0 + b_1 y_2 + b_2 z_1 + b_3 z_2 + u$$
- ◆ We suspect that y_2 is endogenous and we have instruments for y_2 (z_3, z_4)
- ◆ How do we determine if y_2 is endogenous?

Testing for Endogeneity (cont)

1. Hausman Test

- ◆ If all variables are exogenous both OLS and 2SLS are consistent
- ◆ If there are statistically significant differences in the coefficients we conclude that y_2 is endogenous

2. Regression Test

- ◆ In the first stage equation:

$$y_2 = p_0 + p_1 z_1 + p_2 z_2 + p_3 z_3 + p_3 z_3 + v_2$$

- ◆ Each of the z 's are uncorrelated with u_1

Testing for Endogeneity (cont)

- ◆ Thus y_2 will only be correlated with u_1 if v_2 is correlated with u_1
- ◆ The test is based on this observation
- ◆ So, save the residuals from the first stage
- ◆ Include the residual in the structural equation (which of course has y_2 in it)
- ◆ If the coefficient on the residual is statistically different from zero, reject the null of exogeneity
- ◆ If multiple endogenous variables, jointly test the residuals from each first stage

Testing Overidentifying Restrictions

- ◆ How can we determine if we have a good instrument -correlated with y_2 uncorrelated with u ?
- ◆ Easy to test if z is correlated with y_2
- ◆ If there is just one instrument for our endogenous variable, we can't test whether the instrument is uncorrelated with the error (u is unobserved)
- ◆ If we have multiple instruments, it is possible to test the overidentifying restrictions
- ◆ i.e. to see if some of the instruments are correlated with the error

The OverID Test

- ◆ Using our previous example, suppose we have two instruments for y_2 (z_3, z_4)
- ◆ We could estimate our structural model using only z_3 as an instrument, assuming it is uncorrelated with the error, and get the residuals:

$$\hat{u}_1 = y_1 - \hat{b}_0 - \hat{b}_1 y_2 - \hat{b}_2 z_1 - \hat{b}_3 z_2$$

- ◆ Since z_4 hasn't been used we can check whether it is correlated with $\hat{u}_1$
- ◆ If they are correlated z_4 isn't a good instrument

The OverID Test

- ◆ We could do the same for z_3 , **as long as we can assume that z_4 is uncorrelated with u_1**
- ◆ A procedure that allows us to do this is:
 1. Estimate the structural model using IV and obtain the residuals
 2. Regress the residuals on all the exogenous variables and obtain the R^2 to form nR^2
 3. Under the null that all instruments are uncorrelated with the error, $LM \sim \chi_q^2$ where q is the number of “extra” instruments

Testing for Heteroskedasticity

- ◆ When using 2SLS, we need a slight adjustment to the Breusch-Pagan test
- ◆ Get the residuals from the IV estimation
- ◆ Regress these residuals squared on **all** of the exogenous variables in the model (including the instruments)
- ◆ Test for the joint significance
- ◆ Note: there are also robust standard errors in the IV setting

Testing for Serial Correlation

- ◆ Also need a slight adjustment to the test for serial correlation when using 2SLS
- ◆ Re-estimate the structural model by 2SLS, including the lagged residuals, and using the same instruments as originally
- ◆ Test if the coefficient on the lagged residual (ρ) is statistically different than zero
- ◆ Can also correct for serial correlation by doing 2SLS on a quasi-differenced model, using quasi-differenced instruments